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PRACTICAL : 2

AIM 1 : 1. Smart thermostat (Thermostat.java, main in this file)

(a) Create class Thermostat with private String location, private int temperature, private static final int MIN=16 and MAX=30, and private static int activeCount=0.

(b) Constructor(location, startTemp): store location; set temperature to startTemp if it is

within MIN..MAX else 22; increment activeCount.

(c) Constructor(location): chain with this(location, 22).

(d) raise(): if temperature < MAX add 1, else print “Already at maximum (30)”.

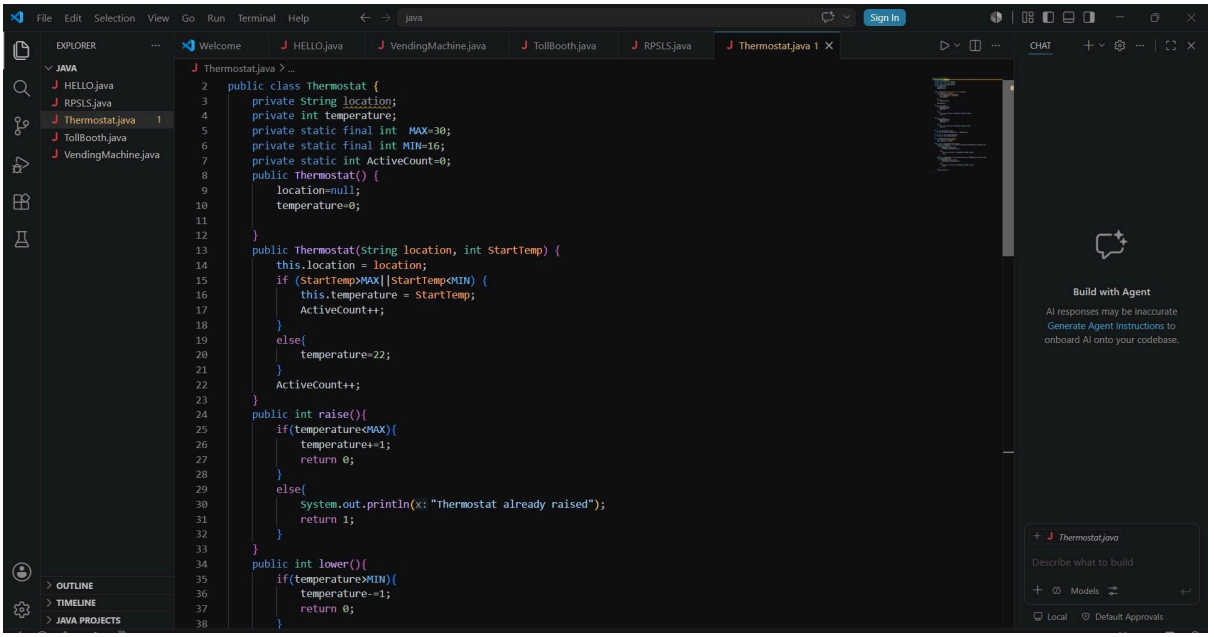
(e) lower(): if temperature > MIN subtract 1, else print “Already at minimum (16)”.

(f) Add getTemperature() (no setter) and static getActiveCount().

(g) In main: create two thermostats, call raise() 10 times then lower() 20 times in loops

(printing the temperature each time), then print getActiveCount().

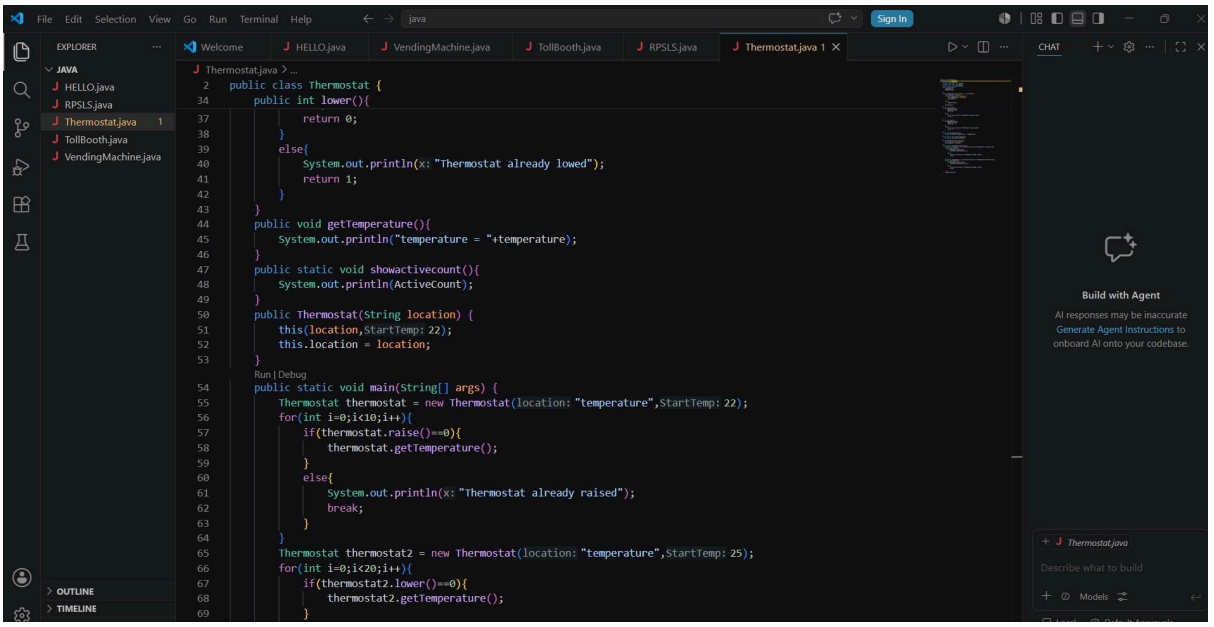
CODE :




```java
public class Thermostat {
    private String location;
    private int temperature;
    private static final int  MAX=30;
    private static final int MIN=16;
    private static int ActiveCount=0;
    public Thermostat() {
        location=null;
        temperature=0;

    }
    public Thermostat(String location, int StartTemp) {
        this.location = location;
        if (StartTemp>MAX||StartTemp<MIN) {
            this.temperature = StartTemp;
            ActiveCount++;
        }
        else{
            temperature=22;
        }
        ActiveCount++;
    }
    public int raise(){
        if(temperature<MAX){
            temperature+=1;
            return 0;
        }
        else{
            System.out.println(x: "Thermostat already raised");
            return 1;
        }
    }
    public int lower(){
        if(temperature>MIN){
            temperature-=1;
            return 0;
        }
```

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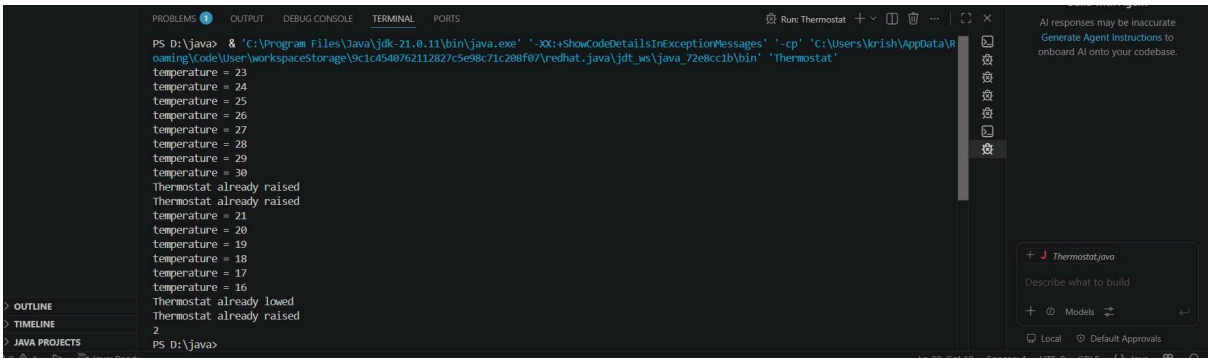


```java
public class Thermostat {
    public int lower(){
        return 0;
    }
    else{
        System.out.println(x: "Thermostat already lowed");
        return 1;
    }
}
public void getTemperature(){
    System.out.println("temperature = "+temperature);
}
public static void showactivecount(){
    System.out.println(ActiveCount);
}
public Thermostat(String location) {
    this(location,StartTemp: 22);
    this.location = location;
}
Run | Debug
public static void main(String[] args) {
    Thermostat thermostat = new Thermostat(location: "temperature",StartTemp: 22);
    for(int i=0;i<10;i++){
        if(thermostat.raise()==0){
            thermostat.getTemperature();
        }
        else{
            System.out.println(x: "Thermostat already raised");
            break;
        }
    }
    Thermostat thermostat2 = new Thermostat(location: "temperature",StartTemp: 25);
    for(int i=0;i<20;i++){
        if(thermostat2.lower()==0){
            thermostat2.getTemperature();
        }
```



```java
    }
    Thermostat thermostat2 = new Thermostat(location: "temperature",StartTemp: 25);
    for(int i=0;i<20;i++){
        if(thermostat2.lower()==0){
            thermostat2.getTemperature();
        }
        else{
            System.out.println(x: "Thermostat already raised");
            break;
        }
    }
    showactivecount();
}
}
```

OUTPUT :



```
PS D:\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\krish\AppData\Roaming\Code\User\workspaceStorage\9c1c4540762112827c5e98c71c208f07\redhat.java\jdt_ws\java_72e8cc1b\bin' 'Thermostat'
temperature = 23
temperature = 24
temperature = 25
temperature = 26
temperature = 27
temperature = 28
temperature = 29
temperature = 30
Thermostat already raised
Thermostat already raised
temperature = 21
temperature = 20
temperature = 19
temperature = 18
temperature = 17
temperature = 16
Thermostat already lowed
Thermostat already raised
2
PS D:\java>
```

AIM 2 : 2. Cinema show (CinemaShow.java, main in this file)

(a) Create class CinemaShow with private String title, private int seatsAvailable, private final int capacity, and private static int totalBooked=0.

(b) Constructor(title, capacity): set both and seatsAvailable = capacity.

Constructor(title): chain with this(title, 100).

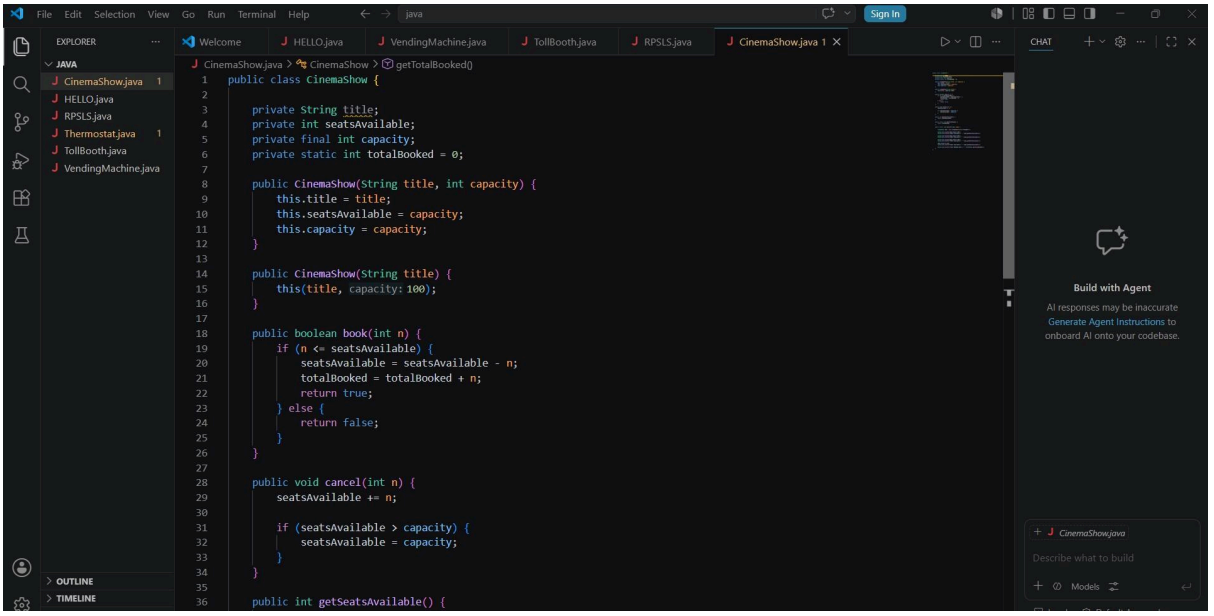
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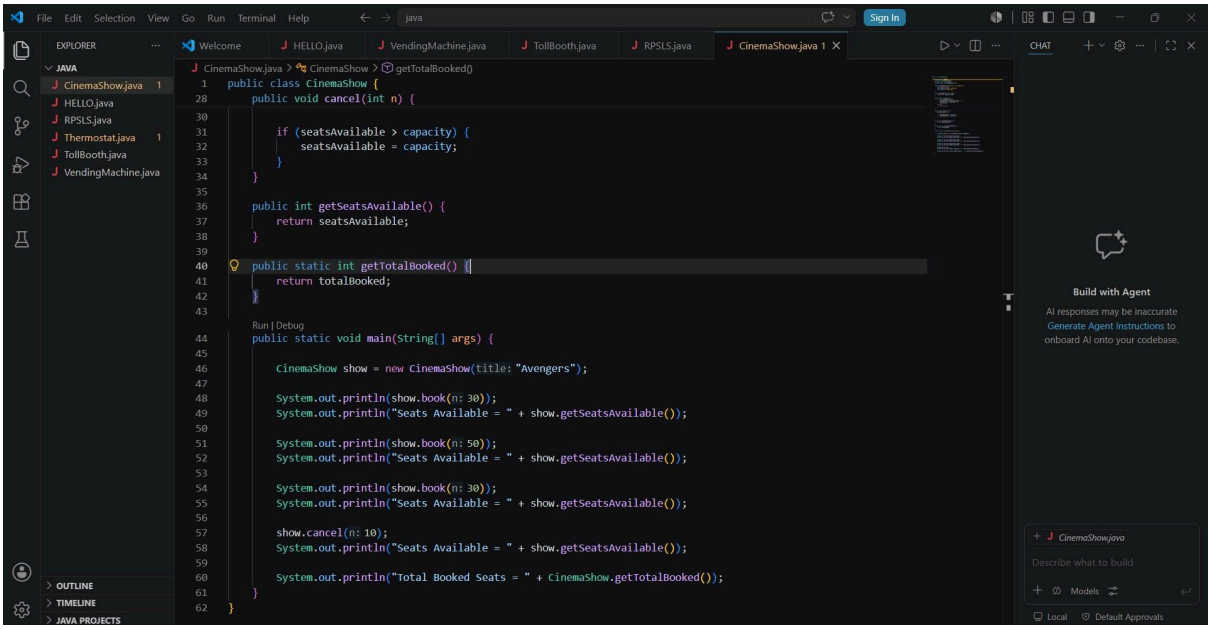
(c) book(int n): if n ≤ seatsAvailable, reduce seatsAvailable by n, add n to totalBooked, return true; else return false unchanged.

(d) cancel(int n): increase seatsAvailable by n but never above capacity.

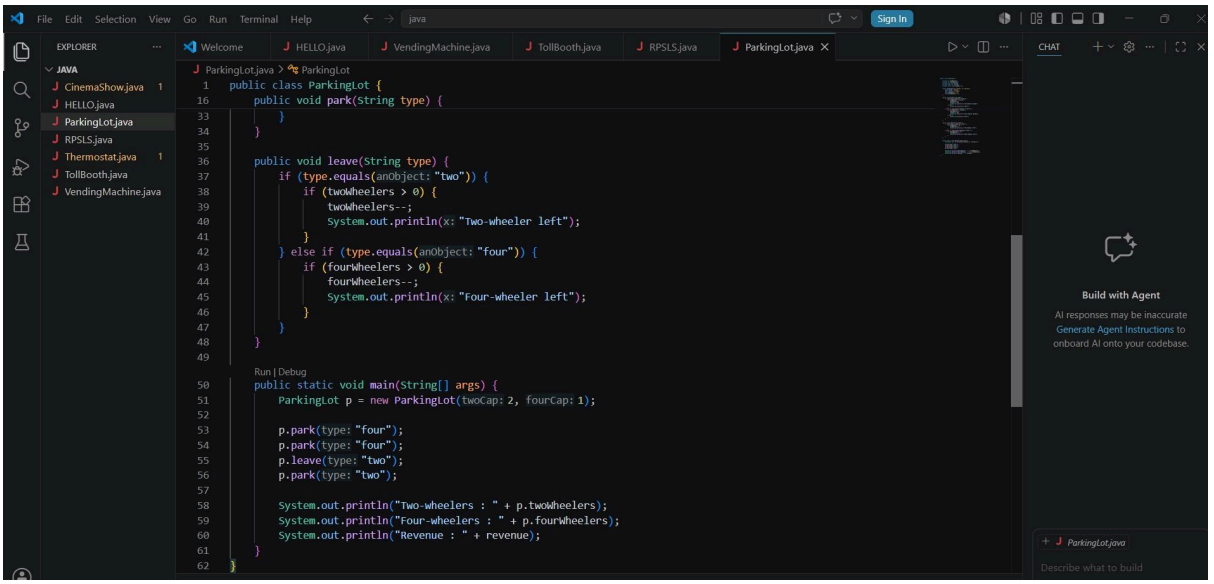
(e) Add getSeatsAvailable() and static getTotalBooked().

(f) In main: run a sequence of book and cancel calls, printing the result and seats after each, then print getTotalBooked().





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```java
public class ParkingLot {
    public void park(String type) {
        }
    }

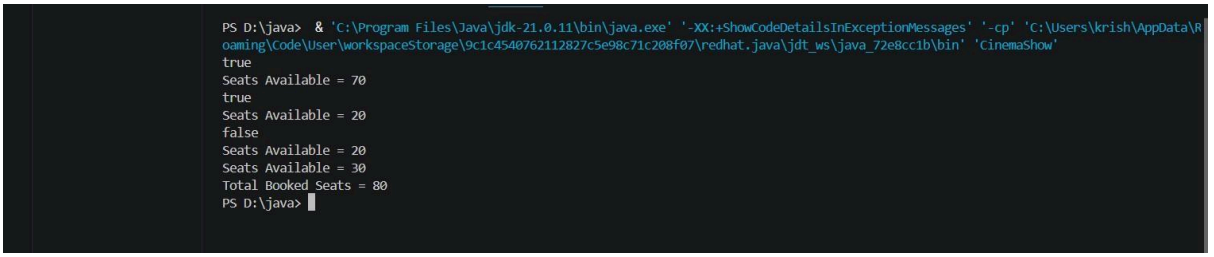
    public void leave(String type) {
        if (type.equals(anObject: "two")) {
            if (twoWheelers > 0) {
                twoWheelers--;
                System.out.println(x: "Two-wheeler left");
            }
        } else if (type.equals(anObject: "four")) {
            if (fourWheelers > 0) {
                fourWheelers--;
                System.out.println(x: "Four-wheeler left");
            }
        }
    }

    Run | Debug
    public static void main(String[] args) {
        ParkingLot p = new ParkingLot(twoCap: 2, fourCap: 1);

        p.park(type: "four");
        p.park(type: "four");
        p.leave(type: "two");
        p.park(type: "two");

        System.out.println("Two-wheelers : " + p.twoWheelers);
        System.out.println("Four-wheelers : " + p.fourWheelers);
        System.out.println("Revenue : " + revenue);
    }
}
```

OUTPUT :



```
PS D:\java>  & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\krish\AppData\R
oaming\Code\User\workspaceStorage\9c1c4540762112827c5e98c71c208f07\redhat.java\jdt_ws\java_72e8cc1b\bin' 'CinemaShow'
true
Seats Available = 70
true
Seats Available = 20
false
Seats Available = 20
Seats Available = 30
Total Booked Seats = 80
PS D:\java>
```

AIM 3 : 3. (For Advanced Learners – home) Parking lot (ParkingLot.java)

(a) Create class ParkingLot with private int twoWheelers and fourWheelers, private final

int twoCap and fourCap, and private static long revenue=0.

(b) park(String type): for “two”, if twoWheelers < twoCap increment and add 20 to revenue; for “four”, if fourWheelers < fourCap increment and add 40; otherwise print

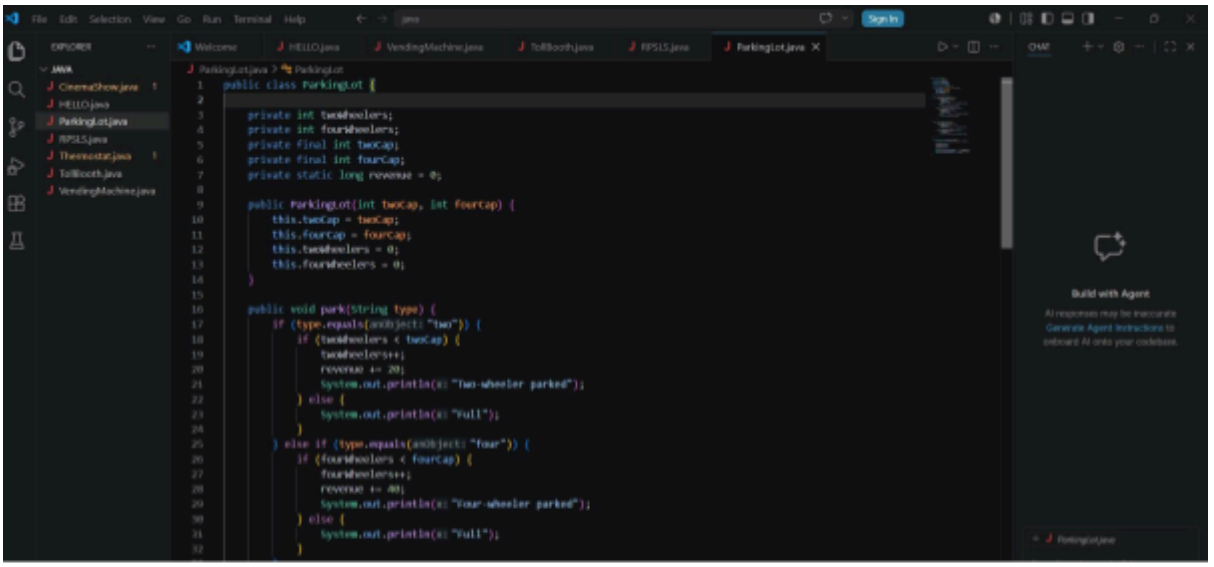
“Full” and reject.

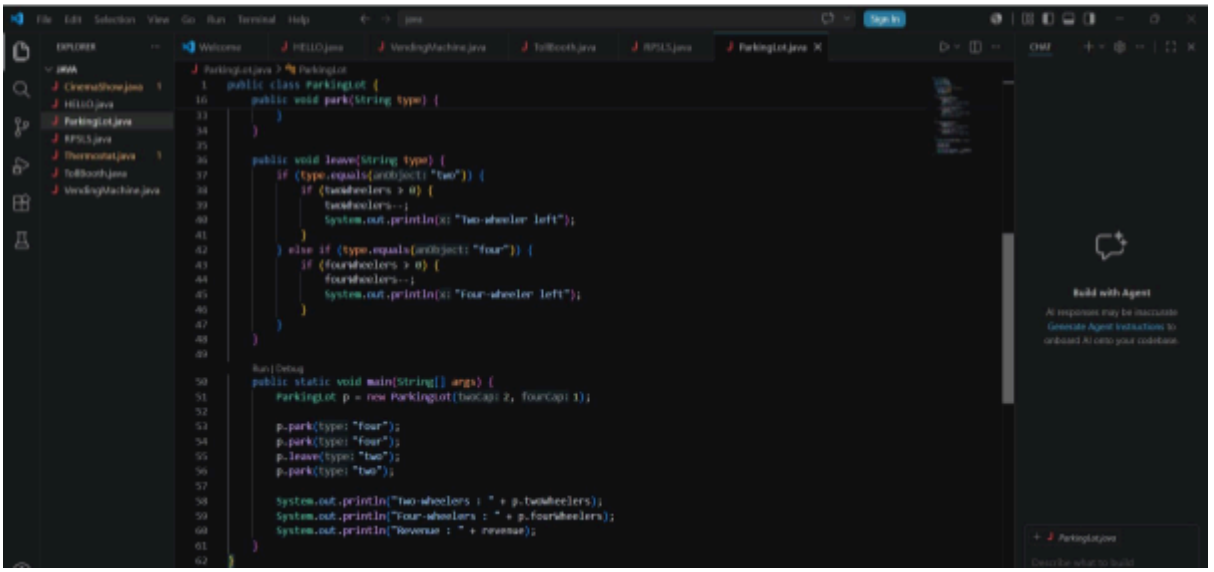
(c) leave(String type): decrement the matching count but never below 0.

(d) In main: simulate a sequence of park/leave events, printing each outcome, then print final occupancy and revenue.

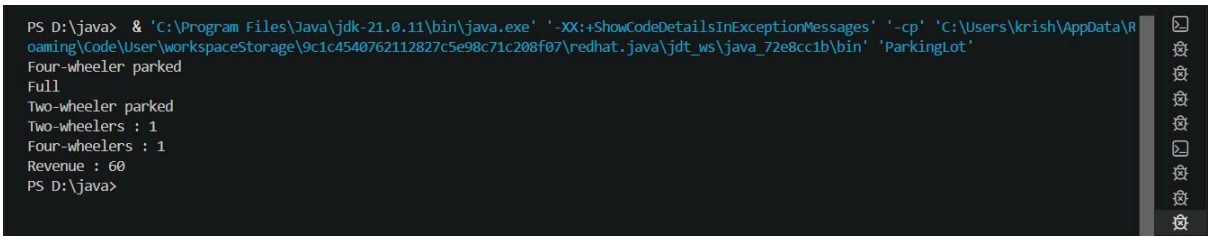
CODE :

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OUTPUT :



```
PS D:\java>  & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\krish\AppData\Roaming\Code\User\workspaceStorage\9c1c4540762112827c5e98c71c208f07\redhat.java\jdt_ws\java_72e8cc1b\bin' 'ParkingLot'
Four-wheeler parked
Full
Two-wheeler parked
Two-wheelers : 1
Four-wheelers : 1
Revenue : 60
PS D:\java>
```

LEARNING OUTCOME :

By completing this practical, the student learns to create Java classes and objects, use constructors, encapsulation, static members, and methods, and apply decision-making and loops to develop simple object-oriented applications. The practical also improves Java programming, debugging, and problem-solving skills.

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